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Efficacy and comparison of antimalarials in cutaneous lupus erythematosus subtypes: a systematic review and meta-analysis*

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Summary

Background The antimalarials (AMs) hydroxychloroquine (HCQ) and chloroquine (CQ) have demonstrated variable cutaneous response rates in cutaneous lupus erythematosus (CLE).
Objectives We sought to assess the global cutaneous response rates to HCQ and CQ, with respect to CLE subtypes, based on previously published studies.
Methods We performed a systematic review and meta-analysis of studies published in MEDLINE, Embase and the Cochrane Library between 1965 and December 2015. The proportions of responders to AMs according to CLE subtypes were extracted from individual studies and pooled using random-effects or fixed models. The odds ratio (OR) was used as the measure of association to compare the response rates between CLE subtypes and AMs.
Results Among 1990 courses of treatment with AMs from 31 included studies, the overall response rate to AMs was 63% [95% confidence interval (CI) 55–70], with important statistical heterogeneity across the included studies. HCQ had a higher overall efficacy than CQ, but this was not significant (OR 1·48, 95% CI 0·98–2·23). The response rate to AMs was different between CLE subtypes, ranging from 31% (95% CI 20–44) for chilblain lupus to 91% (95% CI 87–93) for acute CLE. The response was significantly higher for acute CLE than for subacute CLE and intermittent CLE. In case of failure of monotherapy with AM, the combination of quinacrine with HCQ or CQ seemed effective, whereas too little data were available to assess the efficacy of the switch to another AM agent.
Conclusions Wide discrepancies in cutaneous response to AMs are observed between CLE subtypes. A specific therapeutic approach considering CLE subtypes may improve CLE management.

What's already known about this topic?

- The antimalarials (AMs) hydroxychloroquine (HCQ) and chloroquine (CQ) are usually recommended as the first-line systemic treatment of cutaneous lupus erythematosus (CLE).
- AMs have demonstrated variable cutaneous response rates in CLE.
- Cutaneous response rates to HCQ and CQ, particularly between CLE subtypes, remain largely unknown.

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What does this study add?

- In this meta-analysis, the pooled cutaneous response rate to AMs was 63%, 95% confidence interval (CI) 55–70.
- The response rate to AMs differed between CLE subtypes, ranging from 31% (95% CI 20–44) for chilblain lupus to 91% (95% CI 87–93) for acute CLE.
- The response to AMs was significantly higher for acute CLE than for subacute CLE and intermittent CLE.

Systemic lupus erythematosus (SLE) is an autoimmune disease characterized by the production of autoantibodies to nuclear antigens, and a broad spectrum of clinical manifestations, including specific cutaneous findings in 75–80% of patients.[1] Currently, there is no uniform definition of cutaneous lupus erythematosus (CLE),[2] and different grouping schemes have been discussed based on clinical findings, duration of lesions, pathological findings, direct immunofluorescence results and immunological laboratory abnormalities. At the present time, it is generally accepted[3] that CLE may be divided into acute CLE (ACLE), subacute CLE (SCLE), intermittent CLE (ICLE)/lupus tumidus and chronic CLE (CCLE). The latter is further subdivided into discoid lupus erythematosus (DLE), lupus erythematosus profundus (LEP) and chilblain lupus erythematosus.[3]

First-line treatment of CLE includes general measures such as photoprotection and smoking cessation,[4] as well as local treatments with topical steroids or calcineurin inhibitors,[5] especially in cases of mild CLE. The antimalarials (AMs) hydroxychloroquine (HCQ) and chloroquine (CQ) are usually recommended as the first-line systemic treatment of CLE,[5] and have demonstrated cutaneous response rates of 50–90%.[5] The use of second-line systemic treatments such as the antimalarial quinacrine, or of methotrexate, thalidomide or lenalidomide, is limited by the availability of these drugs in some countries, or by their potentially severe toxicity.[6–9] Therefore, the optimization of AM treatment is an important challenge in patients with CLE. Furthermore, it is important to understand the factors associated with a favourable cutaneous response to AMs. To the best of our knowledge, global cutaneous response rates to HCQ and CQ have never been assessed, particularly between CLE subtypes.

In order to characterize the efficacy profile of AMs better, we performed a systematic review of the literature and a meta-analysis of observational studies or randomized controlled studies to obtain pooled cutaneous response rates to HCQ and CQ in patients with CLE. The primary aim of the study was to assess overall response rates of AM agents and to compare these response rates between CLE subtypes. The secondary aim was to assess the efficacy of the switch of one AM agent to another, as well as that of different AM combinations.

Materials and methods

Literature search and information sources

We searched Embase (1974 to December 2015), MEDLINE (1965 to December 2015) and the Cochrane Database of Systematic Reviews until December 2015 for original articles, without language restrictions. In addition, the grey literature was explored by searching the conference abstracts of the American College of Rheumatology, the American Academy of Dermatology and the European Academy of Dermatology and Venereology until December 2015. The search strategy combined free-text search, exploded MeSH and Emtree terms and all synonyms of the following medical subject headings terms: 'cutaneous lupus erythematosus' (and subtypes), 'antimalarials', 'hydroxychloroquine' and 'chloroquine' (a detailed search strategy is provided in Appendix S1; see Supporting Information). We also searched for additional articles from the reference lists of relevant papers obtained from the electronic search. When the data of interest could not be found in the manuscripts, we contacted the main authors, and excluded the studies for which these data were not available.

Study selection and eligibility criteria

Observational studies or randomized controlled studies were considered if: (i) they included patients with CLE with or without associated SLE; (ii) the number of patients was at least five; (iii) patients received HCQ or CQ as first-line systemic treatment, or a combination therapy (HCQ and quinacrine, or CQ and quinacrine) after failure of a monotherapy with HCQ or CQ; (iv) the number of patients treated with AMs and the number of responders were available to calculate the AM response rate; and (v) data regarding specific assessment of cutaneous involvement were available in the manuscripts. Reviews, editorials, guidelines and case reports were excluded (Fig. 1). The quality of studies was assessed using the Newcastle–Ottawa Scale[10] for observational studies, the Cochrane Collaboration's tool for assessing risk of bias,[11] and the Jadad[12] scale for randomized controlled studies. For the Jadad scale, a score ≥ 3 was considered as adequate trial quality.[13]

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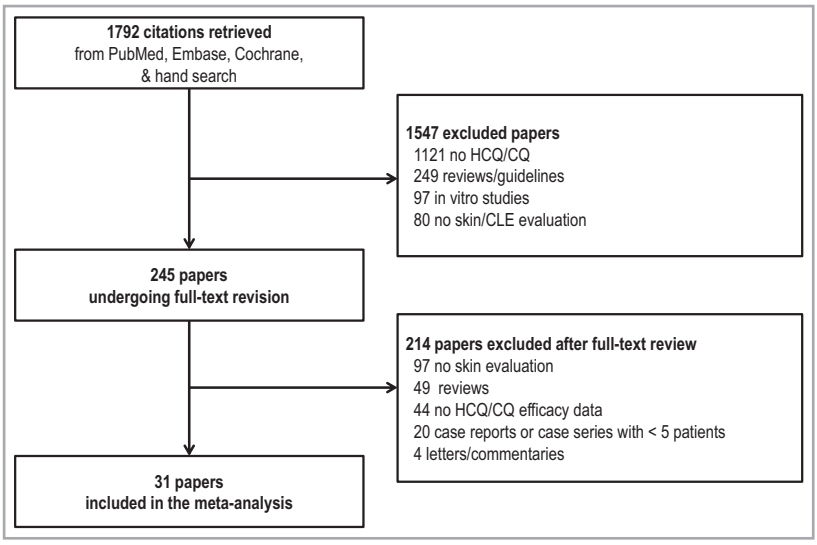

Fig 1. Flowchart for study selection. CLE, cutaneous lupus erythematosus; CQ, chloroquine; HCQ, hydroxychloroquine.

Data extraction

The extracted information for each selected study included the study design, patient characteristics, criteria used for CLE diagnosis, type of AMs used and CLE subtypes. Concurrently, data regarding the efficacy of AMs (or of their combination) were also extracted. These data were extracted independently by two investigators (F.C. and L.A.).

Assessment of cutaneous response to antimicrobials

In some studies, patients may have received more than one AM agent sequentially. Therefore, the total number of courses of use of each AM, rather than the number of patients, was reported. Cutaneous response to AMs was defined according to the definition used in each included study, thus by the Cutaneous Lupus Erythematosus Disease Area and Severity Index (CLASI)[14] or according to study-specific criteria considering the size and number of lesions.

Statistical analyses

The proportion of responders to AMs (HCQ or CQ), as well as according to each CLE subtype, were extracted from individual studies and subsequently pooled following a double-arcsine transformation after continuity correction.[15] In order to compare the efficacy between AMs and CLE subtypes, we constructed 2 × 2 tables and used the odds ratio (OR) as the measure of association. Publication bias was examined using a funnel plot. Heterogeneity (Cochran's χ^2 and I^2 tests) was considered statistically significant at $P < 0{\cdot}10$ and $I^2 > 50\%$. When significant heterogeneity was absent, pooled response rates and 95% confidence intervals (CIs) were calculated using a fixed-effects model, whereas random-effects models according to the method of DerSimonian and Laird[16] were used when significant heterogeneity was present. Subgroup analyses according to study design (prospective vs. retrospective data collection) and the methods to assess cutaneous response (CLASI vs. study-specific response criteria) were performed to analyse potential heterogeneity. The review protocol was planned a priori and is available by contacting the corresponding author. The meta-analysis was performed with the software RevMan version 5·3 (The Cochrane Collaboration, Copenhagen, Denmark) and Meta-XL software version 5·2 (EpiGear International Pty Ltd, Sunrise Beach, Australia).

Results

Literature search and assessment of publication bias

Our literature search identified 1792 citations of interest, from which 31 studies were included in this meta-analysis (Fig. 1).[7,17–46] Visual examination of the funnel plot (Fig. 2) revealed mild asymmetry, suggesting small publication bias and important heterogeneity between studies.

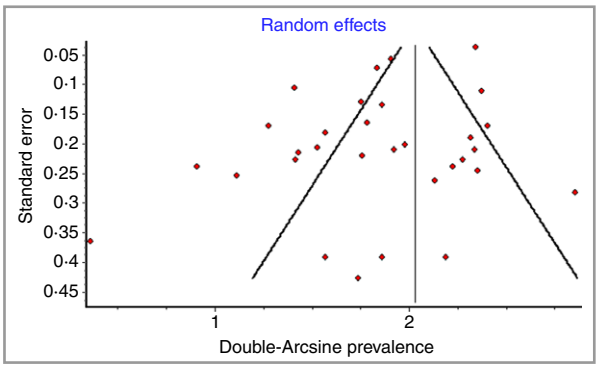

Fig 2. Funnel plot for studies reporting on the efficacy of antimalarials in cutaneous lupus erythematosus. A funnel plot is used to assess publication bias. Red points represent included studies. Visual examination shows slight asymmetry, suggesting small publication bias and heterogeneity.

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Characteristics of the included studies

A total of 26 cohort studies and five randomized controlled trials (RCTs) met the inclusion criteria. The sample size varied from five to 1002 patients with CLE, for a total of 1990 courses of AM use with available data. The mean age at study entry was 40·5 years (range 24·9–56·6), and 76% of patients (range 45–100%) were female. The detailed characteristics of these 31 studies are reported in Table 1. The quality of the studies according to the Newcastle–Ottawa Scale[10] for cohort studies and to both the Cochrane Collaboration's tool for assessing risk of bias[11] and the Jadad scale[12] for RCTs is reported in Appendix S2 (see Supporting Information). For cohort studies, the Newcastle–Ottawa Scale ranged from 3 to 7 stars (out of a maximum of 8 stars). For the RCTs, all five included studies had a Jadad score ≥ 3, suggesting good study quality.

CLE diagnosis was both clinical and histological in 25 studies, histological only in one study, clinical only in one study (defined by compatible cutaneous lesions among patients with SLE with four or more American College of Rheumatology criteria)[47] and unspecified in four studies. Considering AM agents, 13 studies used HCQ only,[17,20–22,27,33,34,37,38,41–43,46] six used CQ only[7,18,25,30–32] and 12 used either one or the other.[19,23,24,26,28,29,35,36,39,40,44,45] Cutaneous response to AMs was assessed by CLASI in seven studies[34,39–44] and according to study-specific criteria in 24 studies.[7,17–33,35–38,45,46] When CLASI was used, cutaneous response was defined by either a 4-point decrease or a 20% decrease in the CLASI activity score in six studies, according to a previously validated definition of improvement,[14] or by a CLASI of 0 or 1 in one study.[34]

Cutaneous response to antimalarials in cutaneous lupus erythematosus

Based on a total of 1990 courses of AMs reported in the 31 included studies,[7,17–46] the pooled response rate to AMs was

Table 1 Characteristics of the included studies

Study	Design	Courses of AMs used	CLE diagnosis	Mean age (years)[a]	Female (%)	AM	Evaluation criteria	Study duration (months)
Kraak 1965[17]	RCT	20	NA	NA	NA	HCQ	Subjective	15
Tuffanelli 1971[18]	Retrospective	6	Clinical + histological	29·5 (12–55)	100	CQ	Subjective	72
Millard 1978[19]	Retrospective	17	Clinical + histological	31·4 (15–56)	100	HCQ/CQ	Subjective	240
Callen 1982[20]	Retrospective	34	Clinical + histological	38·9 (16–73)	66	HCQ	Subjective	60
Spann 1988[21]	Retrospective	7	Clinical + histological	52·3 (34–66)	55	HCQ	Subjective	NA
Ruzicka 1992[22]	RCT	30	NA	44·1 ± 13·3	63	HCQ	Subjective	14
Le Bozec 1994[23]	Retrospective	81	Clinical + histological	34·6 ± 14·2	62	HCQ/CQ	Subjective	120
Rahman 1998[24]	Retrospective	34	Clinical + histological	41·3 (13–67)	77	HCQ/CQ	Subjective	12
Kuhn 2000[25]	Retrospective	22	Clinical + histological	44·8 (10–68)	45	CQ	Subjective	168
Jewell 2000[26]	Retrospective	61	Histological	NA	70	HCQ/CQ	Subjective	48
Versapuech 2000[27]	Retrospective	23	Clinical + histological	41 (23–94)	75	HCQ	Subjective	216
Lardet 2004[28]	Prospective	24	Clinical + histological	35·5 (19–68)	75	HCQ/CQ	Subjective	12
Chieregato 2004[29]	Retrospective	19	NA	55·5 (22–77)	79	NA	Subjective	60
Choonhakarn 2004[30]	Retrospective	14	Clinical + histological	42 (18–64)	67	CQ	Subjective	84
Bezerra 2005[31]	RCT	17	Clinical + histological	34·4 (NA)	94	CQ	Subjective	18
Jacyk 2006[32]	Prospective	6	Clinical + histological	28·5 (10–44)	100	CQ	Subjective	156
Bouaziz 2007[33]	Retrospective	15	Clinical + histological	33 (10–68)	84	HCQ	Subjective	NA
Kreuter 2009[34]	Retrospective	36	Clinical + histological	47·4 (24–70)	58	HCQ	CLASI	24
Cozzani 2010[35]	Retrospective	16	Clinical + histological	43 (19–78)	57	HCQ/CQ	Subjective	NA
Mahfoudh 2010[36]	Retrospective	54	Clinical + histological	42 (12–72)	66	HCQ/CQ	Subjective	12
Wahie 2011[37]	Retrospective	190	Clinical + histological	40 (16–81)	75	HCQ	Subjective	NA
Francès 2012[38]	Prospective	300	Clinical + histological	46·6 (12–85)	84·3	HCQ	Subjective	18
Chang 2011[39]	Prospective	21	Clinical + histological	51 ± 14	86	HCQ/CQ	CLASI	42
Piette 2012[40]	Prospective	88	Clinical + histological	NA	NA	HCQ/CQ	CLASI	43
Ikeda 2012[41]	Prospective	6	NA	45·5 (40–53)	71	HCQ	CLASI	4
Islam 2012[7]	RCT	19	Clinical	24·9 ± 7	96	CQ	Subjective	16
Yokogawa 2013[42]	Prospective	27	Clinical + histological	40·7 (18–58)	85	HCQ	CLASI	NA
Momose 2013[43]	Retrospective	5	Clinical+ histological	38·8 (19–52)	100	HCQ	CLASI	18
Kuhn 2014[44]	Prospective	706	Clinical + histological	50·2 ± 15·4	76·8	HCQ/CQ	CLASI	47
Rodriguez-Caruncho 2015[45]	Retrospective	22	Clinical + histological	37·4 (19–64)	60	NA	Subjective	264
Yokogawa 2015[46]	RCT	72	Clinical + histological	NA	NA	HCQ	Subjective	13

AM, antimalarial; CLASI, Cutaneous Lupus Erythematosus Disease Area and Severity Index; CQ, chloroquine; HCQ, hydroxychloroquine; NA, not available; RCT, randomized controlled trial. [a]Mean ± SD or mean (range) is presented given the available data.

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63% (95% CI 55–70), with significant statistical heterogeneity across the included studies (P < 0·001, I^2 = 89%). For HCQ, 1284 courses of use among 16 studies were included,[17,20–22,26–28,33,37–39,41–44,46] yielding an overall response rate to HCQ of 61% (95% CI 50–71), with significant statistical heterogeneity (P < 0·001, I^2 = 90%). For CQ, 393 courses of use reported in nine studies were included.[7,18,25,28,30–32,36,44] The overall response rate to CQ was 79% (95% CI 75–83), without significant statistical heterogeneity (P = 0·25, I^2 = 21%). Only two studies allowed direct comparisons between HCQ and CQ[28,44] in terms of cutaneous response. In these two studies, the pooled estimate shows a higher efficacy of HCQ than CQ, although this was not significant (OR 1·48, 95% CI 0·98–2·23, P = 0·06), without statistical heterogeneity (P = 0·65, I^2 = 0%) (Fig. S1; see Supporting Information).

Subgroup analyses according to study design showed an overall response rate of 65% (95% CI 50–79) for prospective studies, with statistical heterogeneity (P = 0·01, I^2 = 93%), and for retrospective studies 60% (95% CI 50–70), with statistical heterogeneity (P = 0·01, I^2 = 83%). Moreover, subgroup analyses according to the methods used to assess cutaneous response showed a cutaneous response rate of 64% (95% CI 43–83) with statistical heterogeneity (P = 0·01, I^2 = 92%) for studies using CLASI, and 62% (95% CI 55–70) with statistical heterogeneity (P = 0·01, I^2 = 80%) for studies using study-specific response criteria.

Cutaneous response rates to antimalarials according to cutaneous lupus erythematosus subtypes

The response rates to AMs for the different CLE subtypes are reported in Figure 3.

Based on a total of 872 courses of AMs used in 13 studies,[17,20,21,23,26,28,29,37–39,41,43,44] the overall response rate to AMs in DLE was 57% (95% CI 41–72) (Fig. 4), with significant statistical heterogeneity across the included studies (P < 0·001, I^2 = 94%). For SCLE, eight studies[26–28,38,39,41,43,44] and 279 courses of AM use yielded an overall response rate of 63% (95% CI 42–83), with significant heterogeneity (P < 0·001, I^2 = 88%). For ACLE, a total of 234 courses of AM use reported in two studies[39,44] yielded an overall response rate of 91% (95% CI 87–93), with no heterogeneity (P = 0·88, I^2 = 0%). For ICLE, the overall response rate was 68% (95% CI 56–78) with significant heterogeneity (P = 0·01, I^2 = 61%) based on 145 courses of AM use reported in nine studies.[25,30,34,35,38,39,41,44,45] For chilblain lupus, the cutaneous response was assessed in three studies,[19,33,38] yielding a total of 58 courses of AMs. The overall response rate was 31% (95% CI 20–44) without heterogeneity across the included studies (P = 0·24, I^2 = 30%). Finally, the cutaneous response to AMs for LEP was 48% (95% CI 31–66) with no heterogeneity (P = 0·23, I^2 = 30%), based on four studies[18,32,38,39] that reported on 29 courses of AM use.

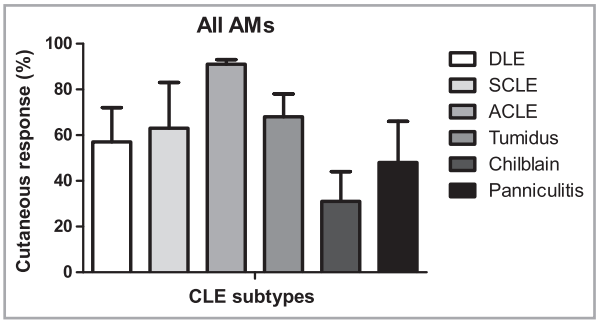

Fig 3. Efficacy of antimalarials (AMs) among cutaneous lupus erythematosus (CLE) subtypes. DLE, discoid lupus erythematosus; SCLE: subacute CLE; ACLE, acute CLE. CLE subtypes are classified in descending order of the number of courses of AMs used.

Comparison of cutaneous response rates to antimalarials between cutaneous lupus erythematosus subtypes

Based on two studies[39,44] with available data, the cutaneous response rate to AMs was higher for ACLE than for SCLE (OR 2·45, 95% CI 1·33–4·52) (Fig. S2a; see Supporting Information) and ICLE (OR 3·90, 95% CI 1·81–8·38) (Fig. S2b), but not statistically different from that of DLE (OR 7·76, 95% CI 0·23–264·48) (Fig. S2c). There were no statistical differences between response to AMs for SCLE compared with DLE (Fig. S3a; see Supporting Information) or ICLE (Fig. S3b), or between DLE and ICLE (Fig. S3c).

Effect of switching antimalarials after failure of first-line treatment

Two studies reported on cutaneous response to CQ after failure of HCQ.[26,39] Of only five patients with available data regarding response to CQ after failure of HCQ, three had an unsustained response after the switch and all five eventually relapsed. In one study,[25] three patients with ICLE refractory to CQ at 3–4 mg kg^{-1} per day were responders to HCQ at 6–6·5 mg kg^{-1} per day.

Cutaneous response rates with the combination of two antimalarials

Four studies reported on a total of 92 courses of combinations of two AMs (HCQ and quinacrine, n = 2; or CQ and quinacrine, n = 2) after failure of monotherapy with AM.[6,39,48,49] The characteristics of these studies are shown in Table 2. The overall response rate for such combinations of AMs was 66% (95% CI 51–80), with significant statistical heterogeneity (P = 0·1, I^2 = 53%) (Fig. S4a; see Supporting Information). Based on the available data, the response rates to these combinations were not different between DLE and SCLE (OR 0·81, 95% CI 0·27–2·37) (Fig. S4b).

Ophthalmological adverse events were assessed in three of these studies. Feldmann *et al.*[48] observed a case of retinopathy after 3 months of treatment, whereas Lipsker *et al.* reported three cases of retinopathy after 1, 3 and 15 months,

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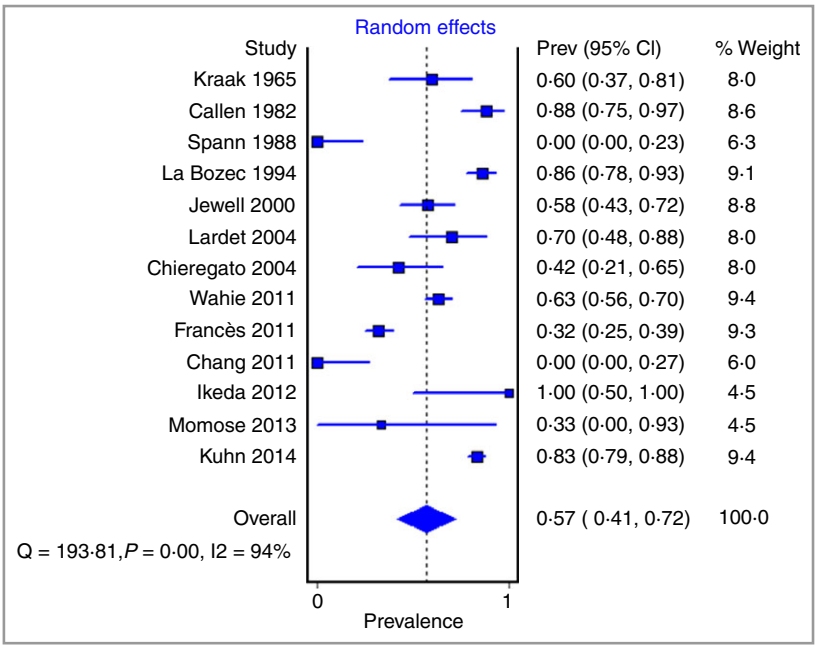

Fig 4. Forest plot showing overall response rates to antimalarials in discoid lupus erythematosus. CI, confidence interval.

Table 2 Characteristics of included studies assessing antimalarial (AM) combinations

Study	Feldmann 1994[48]	Lipsker 1995[49]	Cavazzana 2009[6]	Chang 2011[39]
Design	Retrospective	Retrospective	Prospective	Prospective
Courses of AMs	14	15	42	21
Mean age (years)[a]	44 (24–71)	39 (25–58)	44 ± 13	48 ± 12
Female (%)	57	93	76	66
AM combinations (dose in mg per day)	CQ (300) + Qn (65 × 3)	CQ (300) + Qn (100 × 2)	HCQ (5 mg kg^{-1}) + Qn (50/100)	HCQ (200/400) + Qn (100)
Mean treatment duration (months)	10·7	6·3	5·5	9
Comments	One case of retinopathy after 3 months	Three cases of retinopathy after 1, 5 and 15 months	No retinopathy reported; 100 mg per day more efficient than 50 mg per day but more side-effects	One of five responded to CQ + Qn after failure of HCQ + Qn

CQ, chloroquine; HCQ, hydroxychloroquine; Qn, quinacrine. [a]Mean ± SD or mean (range) is presented given the available data

respectively.[49] Cavazzana *et al.* did not report any retinopathy in a prospective study of 42 patients.[6]

Discussion

In this systematic review of the literature and meta-analysis, we found that the pooled response rate to AMs was 63% (95% CI 55–70). Optimization of AM treatment therefore remains an important challenge in CLE. Our study further revealed major differences in cutaneous response rates to AMs between CLE subtypes, ranging from 31% (95% CI 20–44) for chilblain lupus to 91% (95% CI 87–93) for ACLE.

Which AM is most effective as a first-line treatment in CLE remains controversial. By pooling studies reporting on the efficacy of either HCQ or CQ, we found that the overall cutaneous response rate was higher for CQ. However, pooled estimates from papers that enabled direct comparison between HCQ and CQ suggested that HCQ was more efficacious than CQ, but this was not significant, possibly due to lack of power. These discrepancies may also be due to differences in sample sizes, to the larger reported number of patients treated with HCQ compared with CQ, possibly to differences in the proportion of each CLE subtype, and to the various methods used to assess cutaneous response across the studies.

We found that the response rate of ACLE to AMs was 2·5-fold higher than that of SCLE, and 3·9-fold higher than that of ICLE. While this result reveals significant differences in the efficacy profile of AMs between CLE subtypes, it may also be due to the role of concurrent systemic corticosteroids and other immunosuppressive agents, as those are more

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commonly prescribed in ACLE than in any other CLE subtype, given the frequency of systemic manifestations associated with ACLE.[50] On the contrary, we observed no significant difference in the response to AMs between patients with DLE and SCLE or ICLE, which contradicts the previous observations by Francès *et al.*[38] that DLE has a lower response rate to HCQ than other CLE subtypes. This apparent discrepancy may be due to differences in the proportion of the various forms of DLE or of its extent and severity between studies. Indeed, Spann *et al.* have shown that hypertrophic DLE was resistant to AMs,[21] while Wahie *et al.* reported that disseminated DLE and scarring alopecia from a patient with DLE were associated with the failure of AMs.[37]

There are various strategies that might be used following failure of AM monotherapy. These include smoking cessation,[51] increasing HCQ doses to reach blood concentrations > 750 ng mL^{-1},[52] switching to another AM agent (which seemed poorly effective in the long term, although too few data were available to draw final conclusion)[39] or combining AMs. Regarding the latter strategy, we found a pooled response rate of 66% (95% CI 51–80) for the combination of quinacrine with HCQ or CQ. However, quinacrine is not available in many countries, and may have side-effects such as jaundice-like discoloration of the skin.[48] Solid data regarding the risk of retinal toxicity using combinations of AMs are still missing.

Among the limitations of this meta-analysis is the important statistical heterogeneity measured across the included studies. This might reflect differences between study design, CLE subtypes, doses and types of AM agents, and modalities of cutaneous response assessment. However, subgroup analyses showed similar response rates according to study design and methods used to assess cutaneous response. Consequently, pooled estimates were obtained by combining individual study estimates using random-effects models, as recommended in the context of heterogeneity.[15] In some studies, some patients with more than one CLE subtype were considered to have only the subtype with the highest risk of developing systemic organ manifestations (ACLE > SCLE > CCLE > ICLE).[44] This constraint may have contributed to limiting the sample size in some of the less common CLE subtypes. However, this study allows us to obtain data on rare CLE subtypes. Although direct comparisons with other CLE subtypes were not possible, both chilblain lupus and LEP seemed to be more refractory to AMs than other CLE subtypes, with overall response rates of 31% (95% CI 20–44) and 48% (95% CI 31–66), respectively. Studies focusing on the treatment of these rare CLE subtypes are currently lacking to improve CLE management.

Overall, in this systematic review and meta-analysis of 31 studies including 1990 courses of AM use, we found an overall response rate to AMs of 63% (95% CI 55–70), with important differences between CLE subtypes. While the currently available data did not allow us to draw any formal conclusion regarding the superiority of one AM agent over another, we believe that HCQ should be preferred as the first-line systemic agent, given its better tolerance profile (particularly for retinal toxicity)[53] and the possibility of measuring blood concentration, which is associated with cutaneous response and can be used to identify nonadherent patients.[38,52] In case of refractory CLE, smoking cessation,[51] adaptation of the dose of HCQ to its blood concentration[52] or addition of quinacrine, when available, to HCQ or CQ should be considered. In order to improve CLE management, future prospective studies should focus on various CLE subtypes with standardized outcome reporting measures, in order to define therapeutic strategies based on CLE subtypes.

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Appendix S1. Search terms for PubMed and MEDLINE.

Appendix S2. Assessment of the methodological quality of the included observational studies.

Fig S1. Forest plot showing comparisons of antimicrobial agent efficacy between hydroxychloroquine and chloroquine.

Fig S2. Forest plot showing comparisons of antimicrobial agent efficacy between acute cutaneous lupus erythematosus and other cutaneous lupus erythematosus subtypes.

Fig S3. Forest plot showing comparisons of antimicrobial agent efficacy between cutaneous lupus erythematosus subtypes.

Fig S4. (a) Forest plot showing the overall response rate to antimalarial combinations in cases of refractory cutaneous lupus erythematosus. (b) Forrest plot showing comparison of antimalarial combinations between discoid lupus erythematosus and subacute cutaneous lupus erythematosus.

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